

Artificial Intelligence for Business Value

What is this talk about? (and what it is not!)

What is this talk about:

- Understand how business decisions are made and the relevance of AI / ML
- Understand the basics of Artificial Intelligence and its relevance in business context
- Discuss industry applications of AI / ML

What this talk is not:

- Does not deal with cost-benefit analysis of use cases
- Does not cover moral, ethical dimensions of applications
- Does not cover any math behind the techniques

How delivered: I am going to put myself in your shoes, ask & answer key questions that you might have in your mind as you embark on this course!

Q1: What is the motivation for us to understand Data
Science, AI & ML?



AI/ML is all around us: Movies at the Theatre (An Example)

Recommended for you



Open a movie app to
book a ticket

Watch a movie in
theatre

Snacks during the
break

Write a review about
the movie

Recommendation
Engine based on
Collaborative Filtering

Self driving
car

Predicting box office
sales based on star
value, buzz etc.

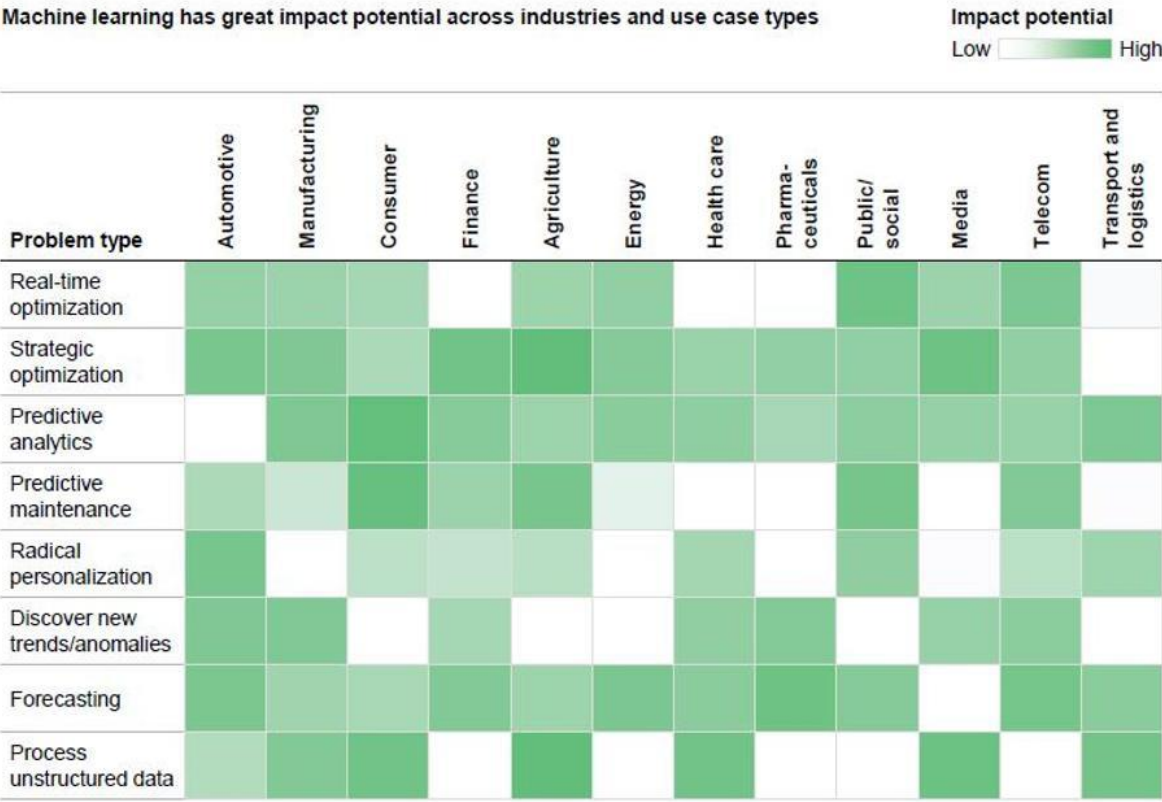
Forecasting product
sales based on
historical trends

Text Analytics to
identify sentiments,
emotions, likes &
dislikes etc.

AI / ML Applications

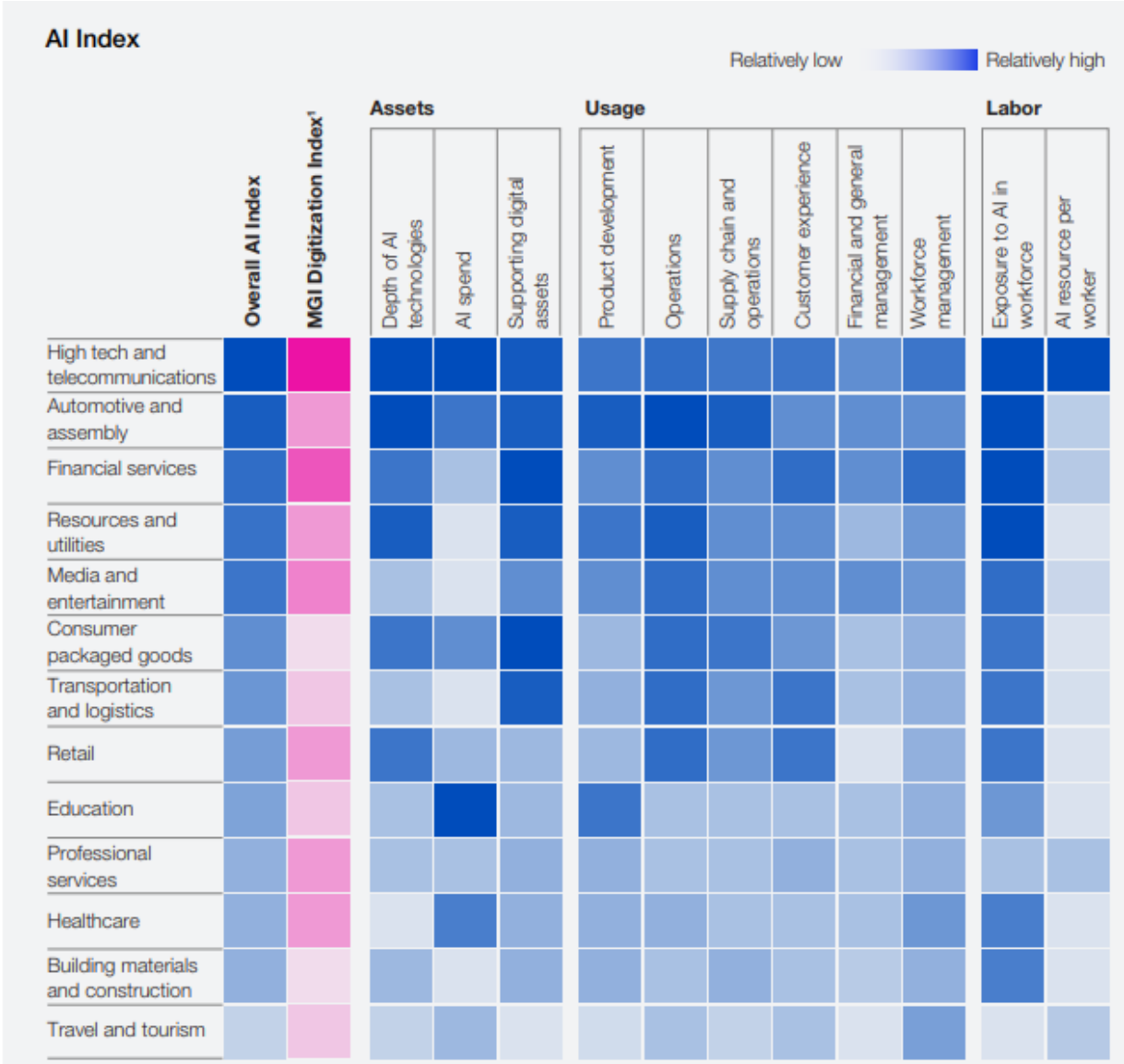
This file is meant for personal use by somureddy@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

There is increasing use of AI/ML across industries = Rewarding Career



SOURCE: McKinsey Global Institute analysis

<https://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/an-executives-guide-to-ai>



But getting true business value needs (lot) more work...

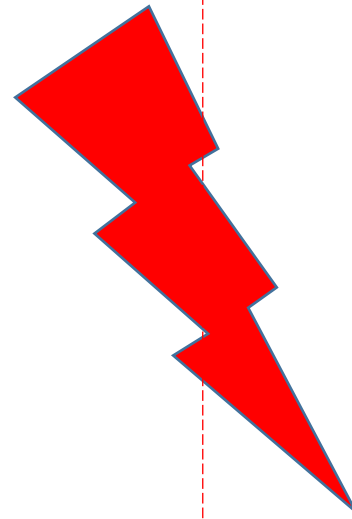
Stacked Ensemble
has an accuracy of
97% without
overfitting

DL Network with 25
hidden layers beat
the existing
benchmarks



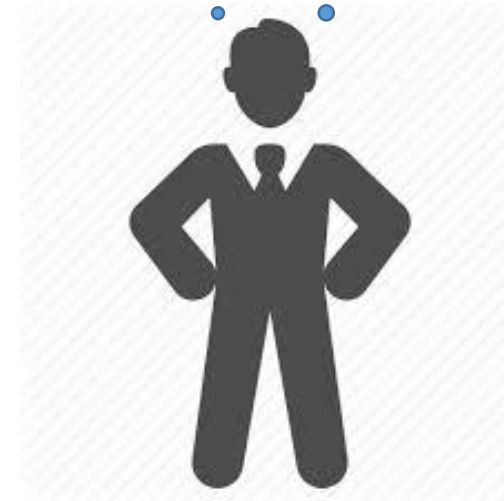
Data Scientists

**Data Science (AI/ML) is powerful
but there exists a divide**



Business Value??
Not so sure

Do they understand
the business
complexity, the
context?

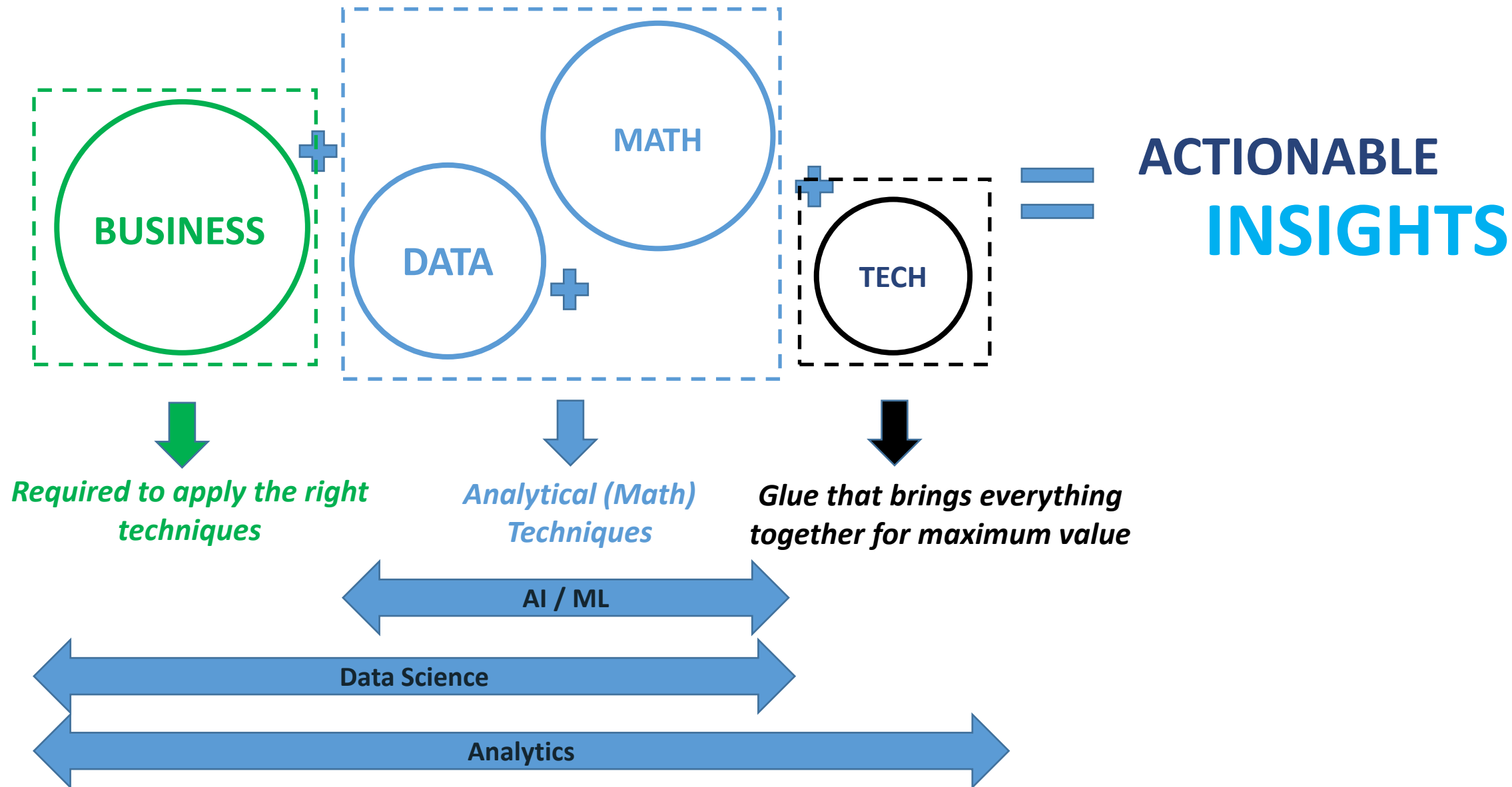


Business Decision Maker

Q2: Before getting into the details, can you show me the simplest possible picture to understand Analytics and its relationship to AI / ML?



Analytics is a tool to solve business problems...



Q3. To start with, can we understand what is Artificial Intelligence? How are AI & ML related?



First, let's try to understand categories of Human Intelligence



Intelligence exhibited by this person

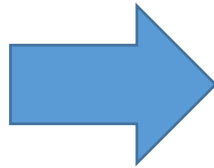
1. Perceive the world, detect signals and collect data
2. Make sense of the world using data (Inference)
3. Decide on the next course of action
4. Act in the Real World

What is Artificial Intelligence?

Artificial Intelligence refers to the theory and development of computer systems & machines with the ability to perform tasks normally requiring human intelligence

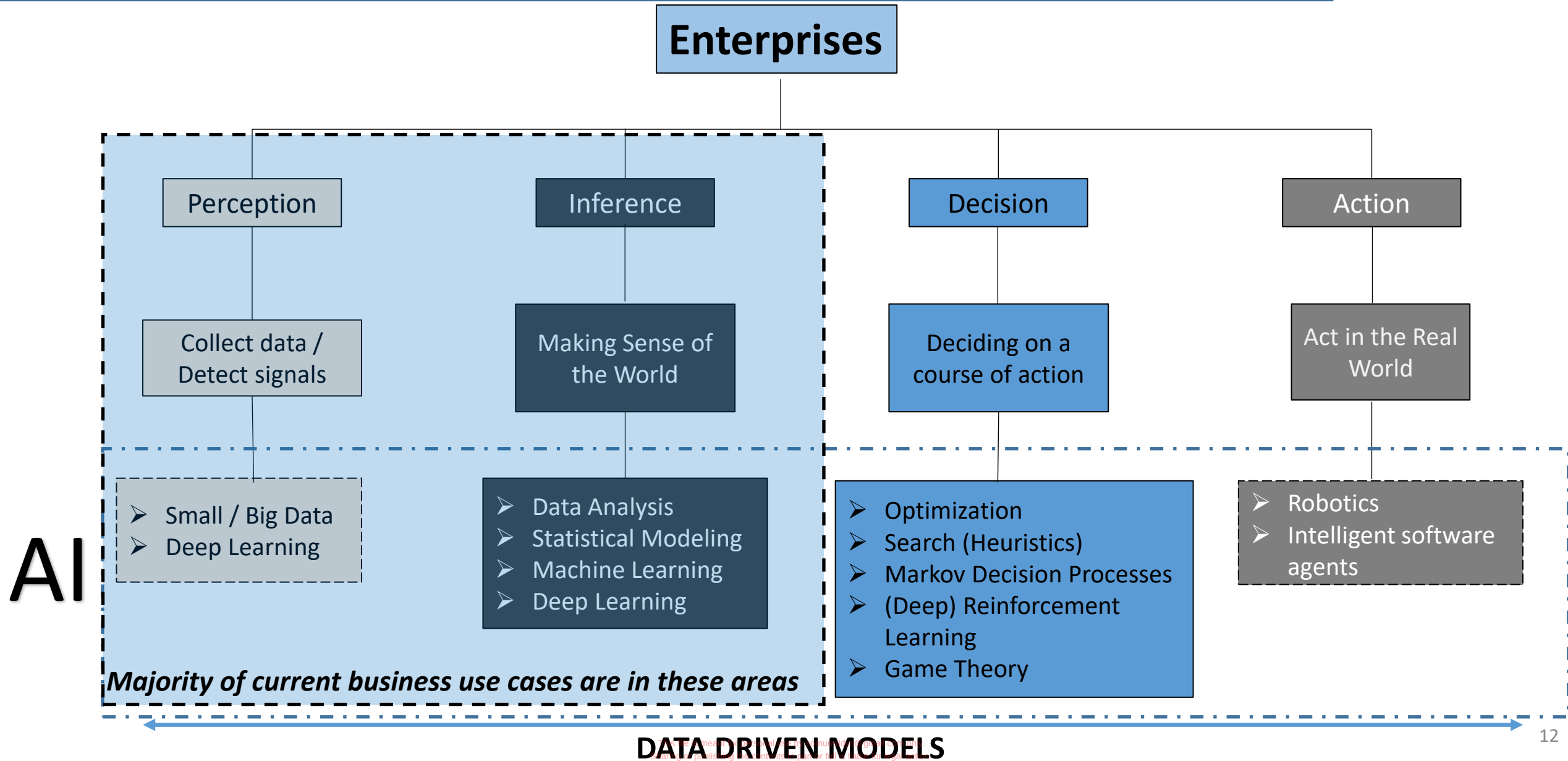
Main Categories of Human Intelligence

1. PERCEPTION
2. INFERENCE
3. DECISION
4. ACTION

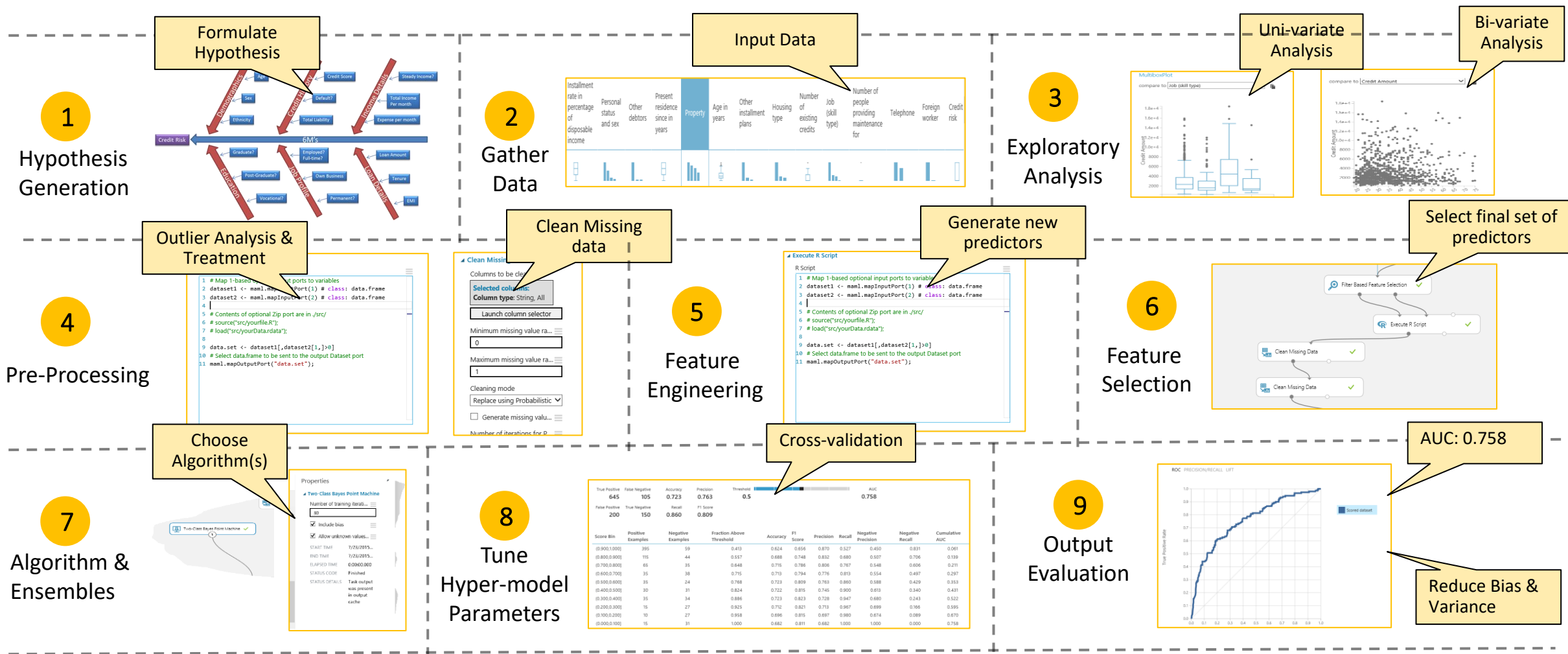


We get our computers to do any or some or all of these tasks by embedding AI into our software programs

Why is AI relevant for business enterprises?



Machine Learning Pipeline – Subset of AI



MANY USE CASES IN THE INDUSTRY ARE BASED ON THIS PIPELINE...

Q4: What are the broad categories of decisions taken in organizations and how are these decisions made?



On a daily basis organizations take hundreds of decisions...

Should I acquire Company A?

What revenue guidance should I give my investors?

What price should I set for my product?

How do I increase checkouts on my ecommerce website?

What campaigns should I run? How much should I spend on those campaigns?

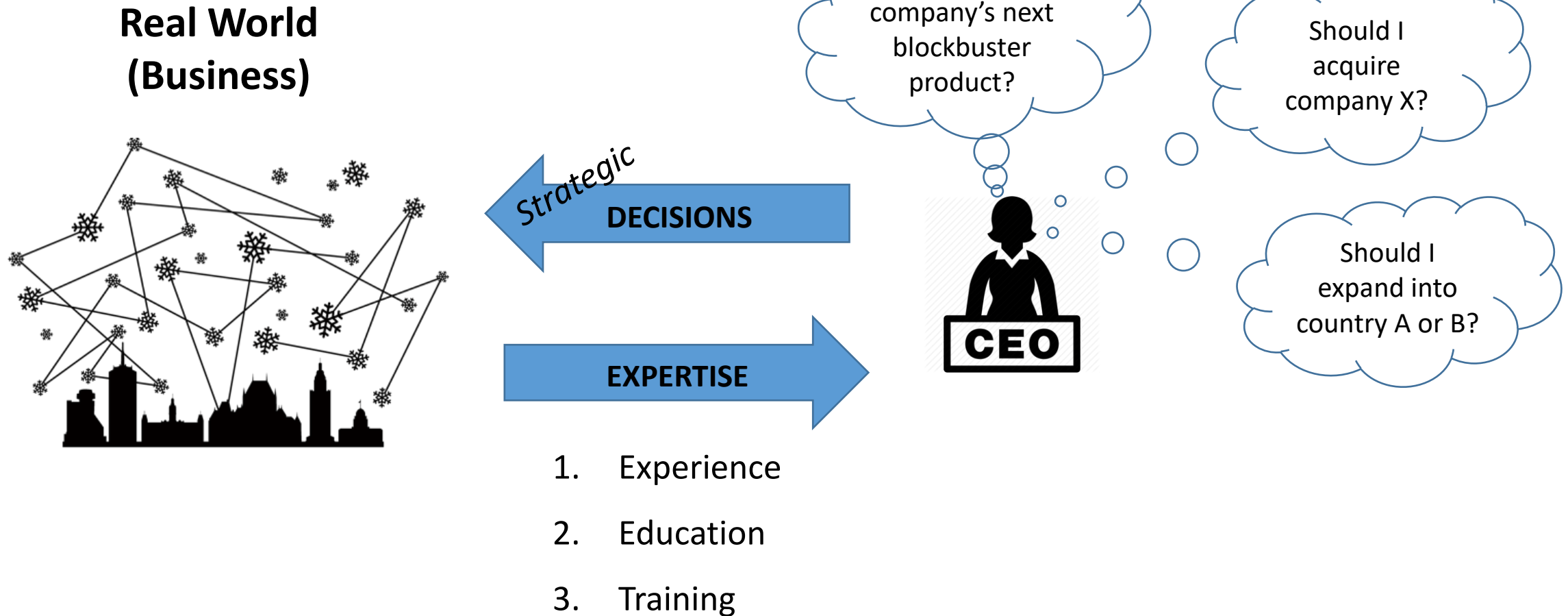
How do I increase manufacturing throughput?

Should I hire this person or not?

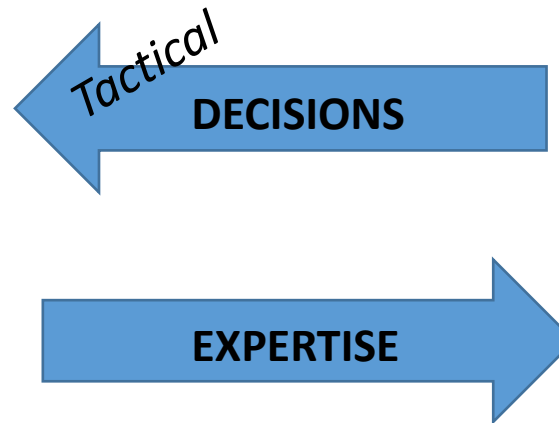
What offers should I give my customers?

How do I comply with regulatory requirements?

Business Decisions – Type 1 - Strategic



Business Decisions – Type 2 - Tactical



1. Education
2. Experience
3. Training



***Head of Sales
& Marketing***

What price
should I set for
Product X, SKU Y?

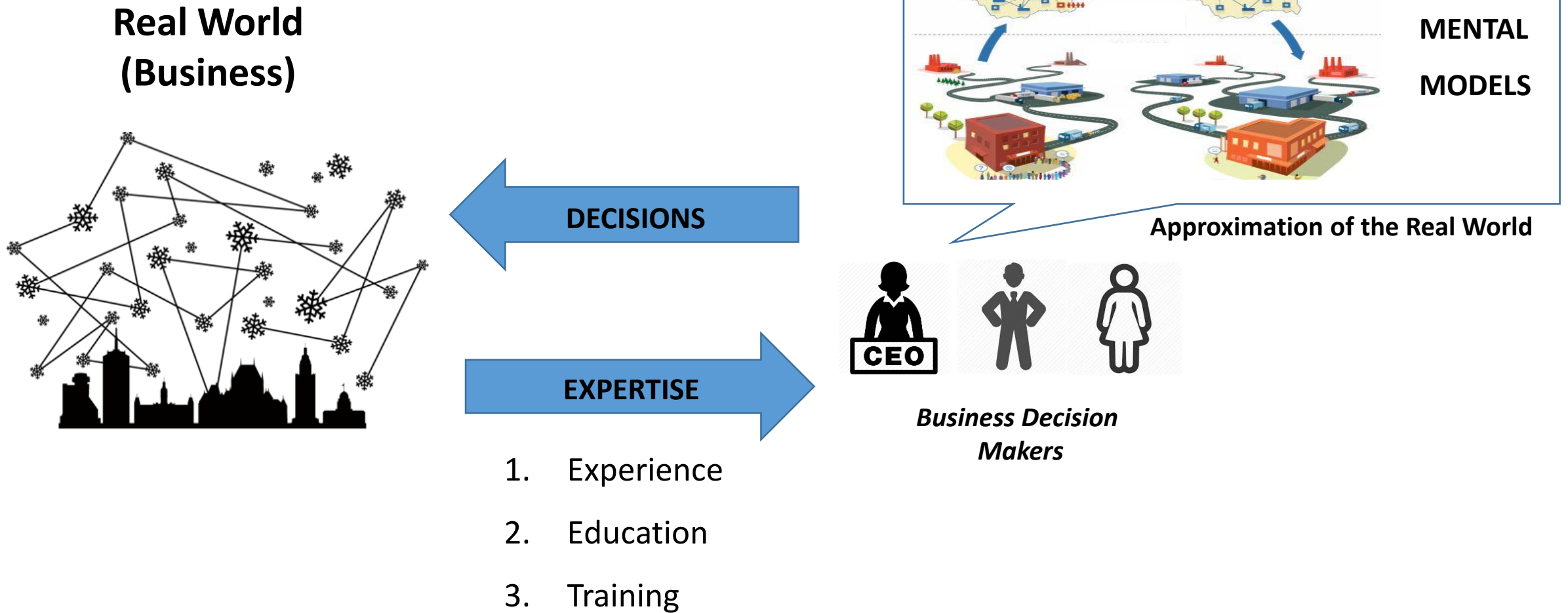
What should the
next week
marketing
campaign focus
on?

How do I handle
the negative
brand perception
on social media?

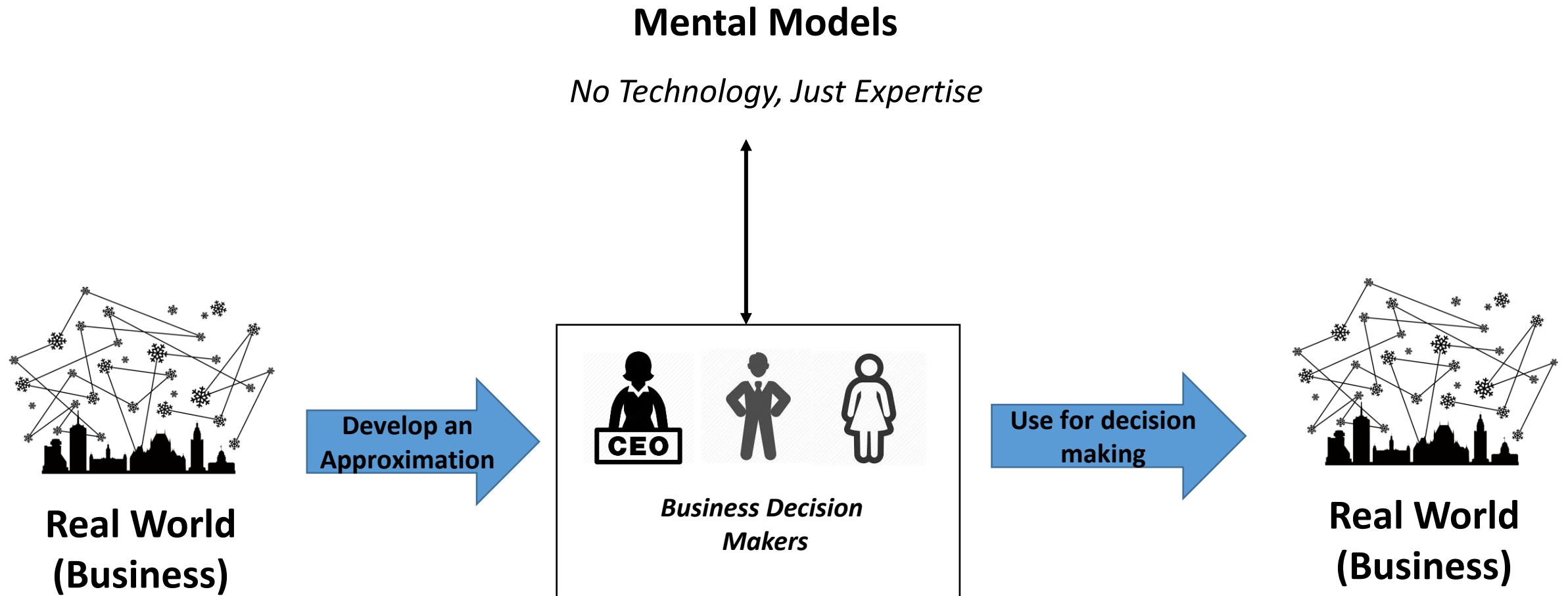
Business Decisions – Type 3 - Operational



So how are business decisions made?



Mental Models backed by Human Expertise (Gut Feel)

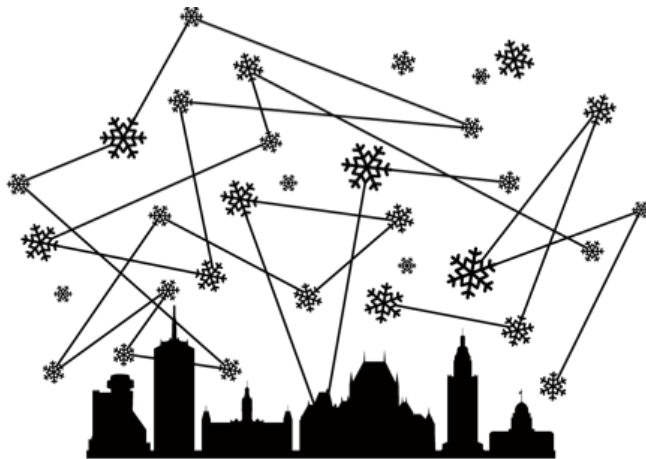


Q5: Decision making in a business context has been done for decades. Where does Analytics (AI/ML) play a role and why is there such a big interest now?



Business Decision Making is becoming increasingly 'Complex'

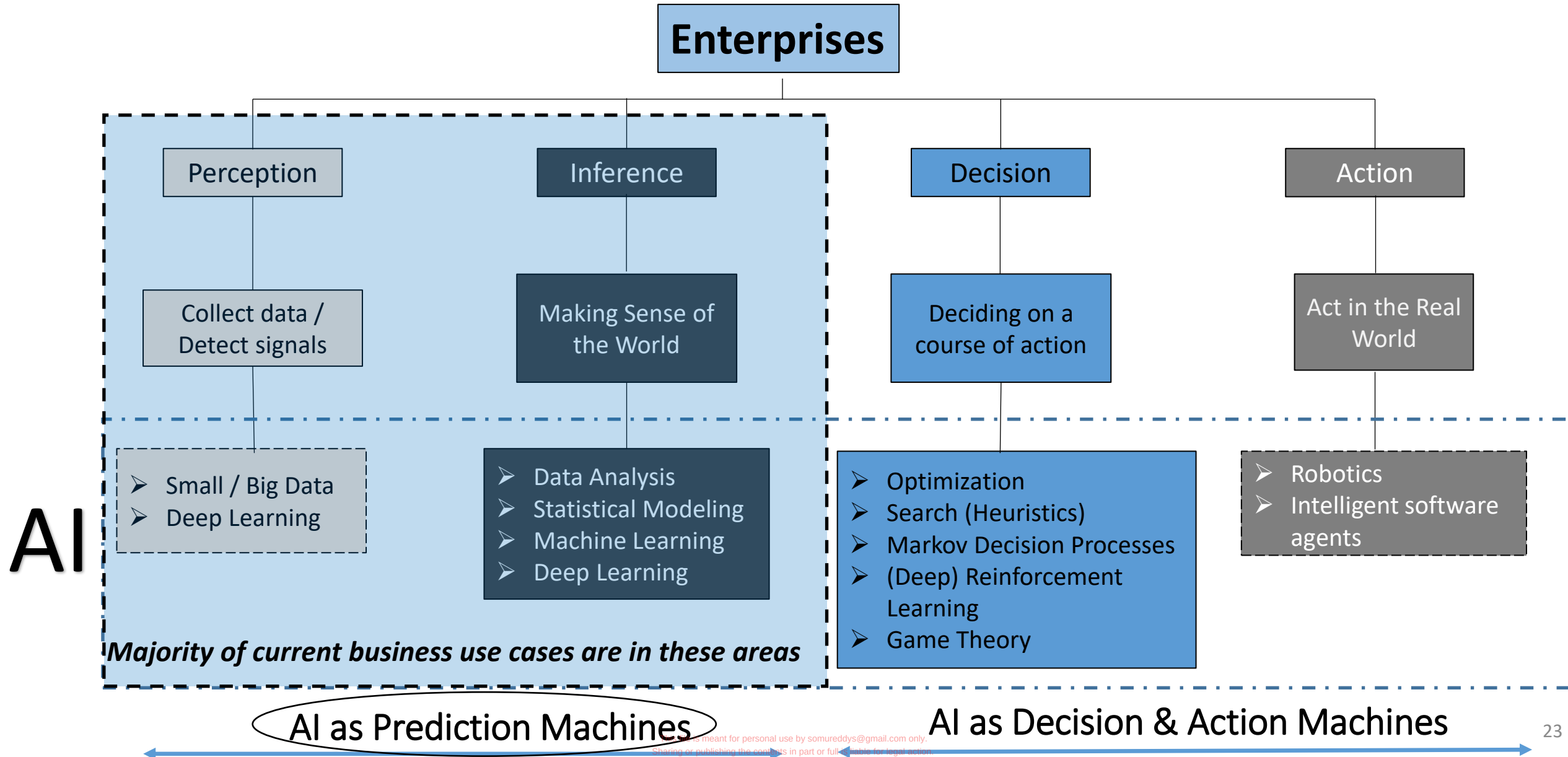
Real World (Business)



- Multiple Entities with their own goals
- Complex Non-Linear Relationships
- Delayed Feedback
- 'Black Swan' Events (Unpredictable events)
- Uncertainty & Long Term Consequences

Business decision makers are expected to take right decisions in this complex world

AI as Prediction Machines



AI as Prediction Machines - Examples

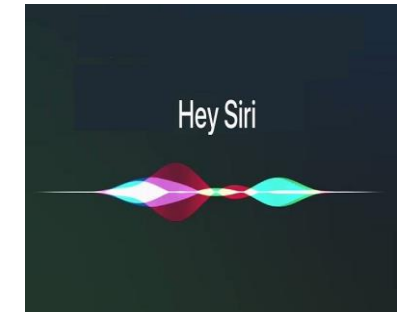
Prediction Machines



Customer Attributes
Product Details
Transaction History

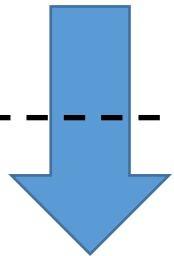


Product Sales
across time
Location of Stores



Utterance by a
customer on his /
her mobile phone

Take Information
that you have
(Data)



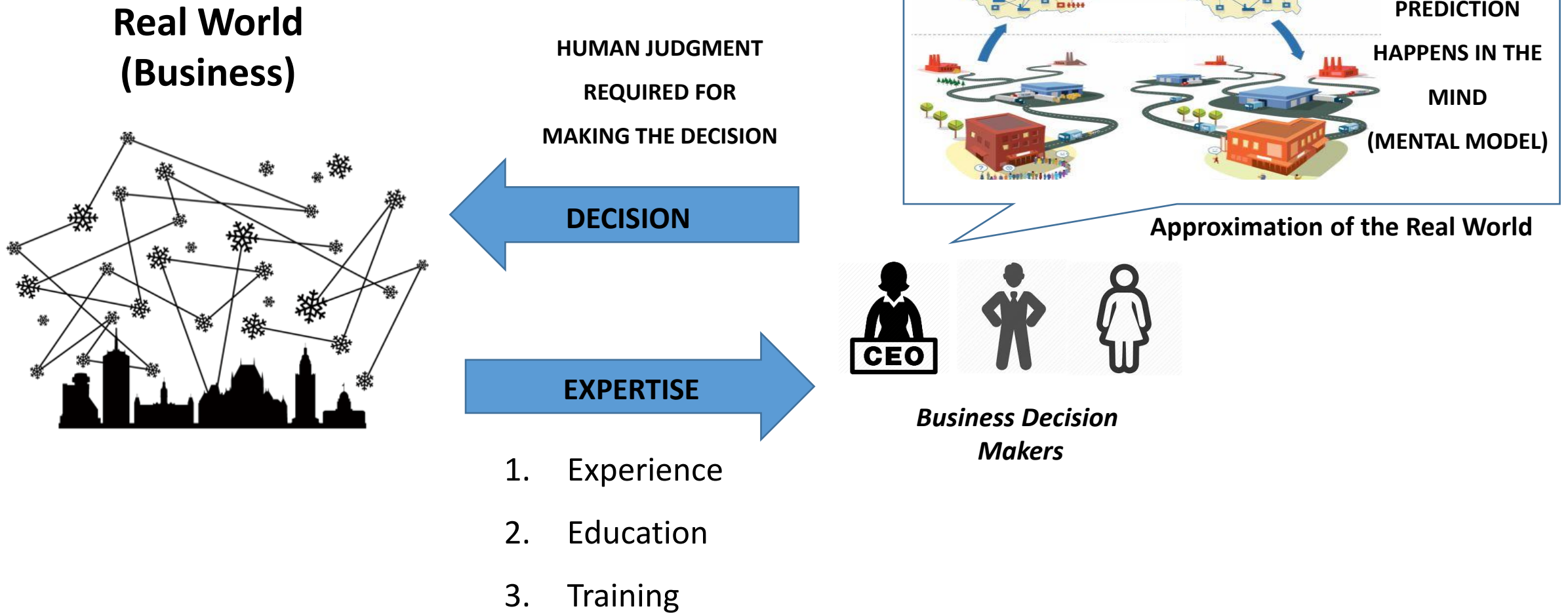
Create Information
that you don't have
(Predictions)

Whether a particular
customer will churn
(leave the bank) or
not?

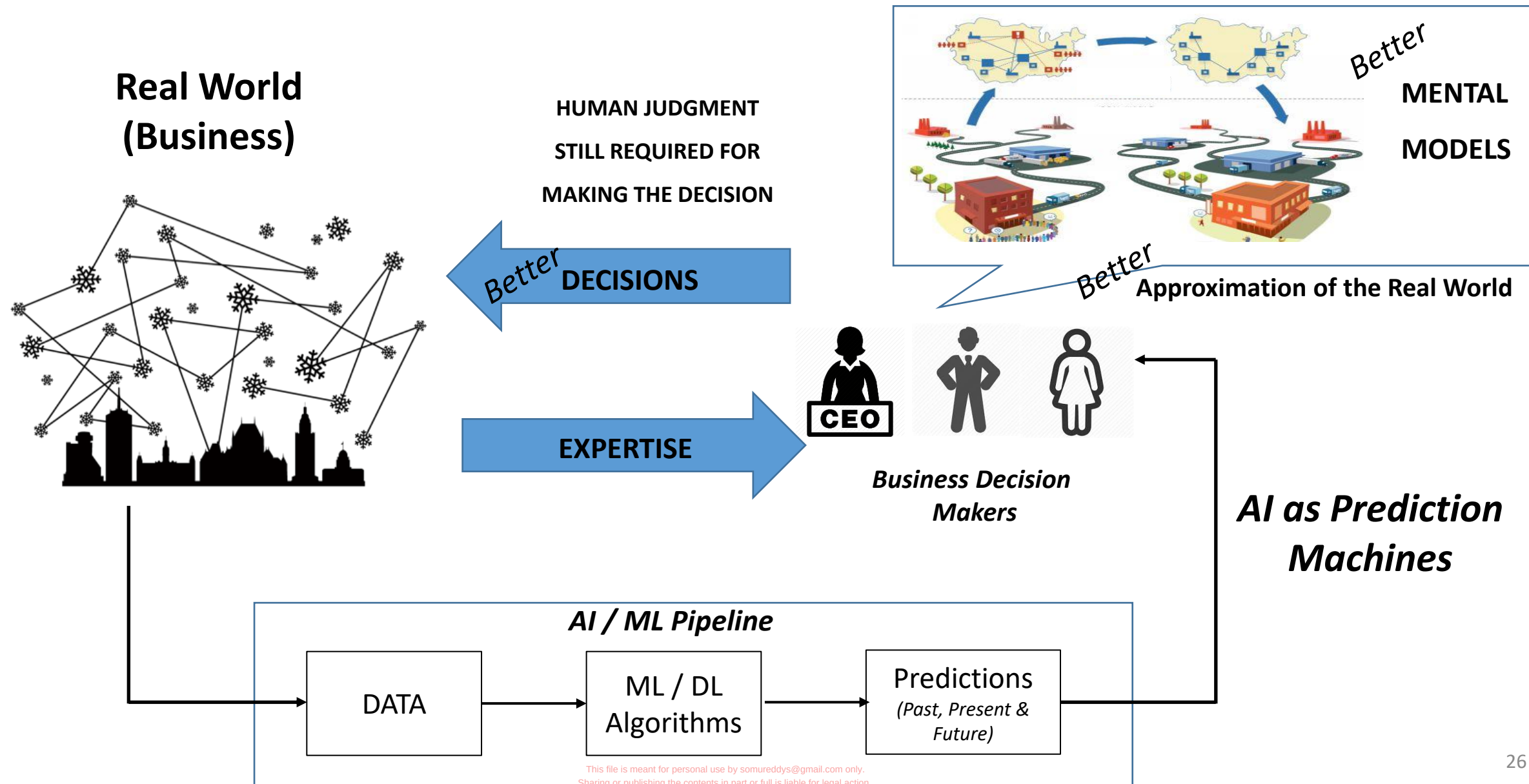
Forecast demand for
products in the next 3
months in each store

Precisely identify
what the customer
said so that action
can be performed

We saw earlier how business decisions are made?

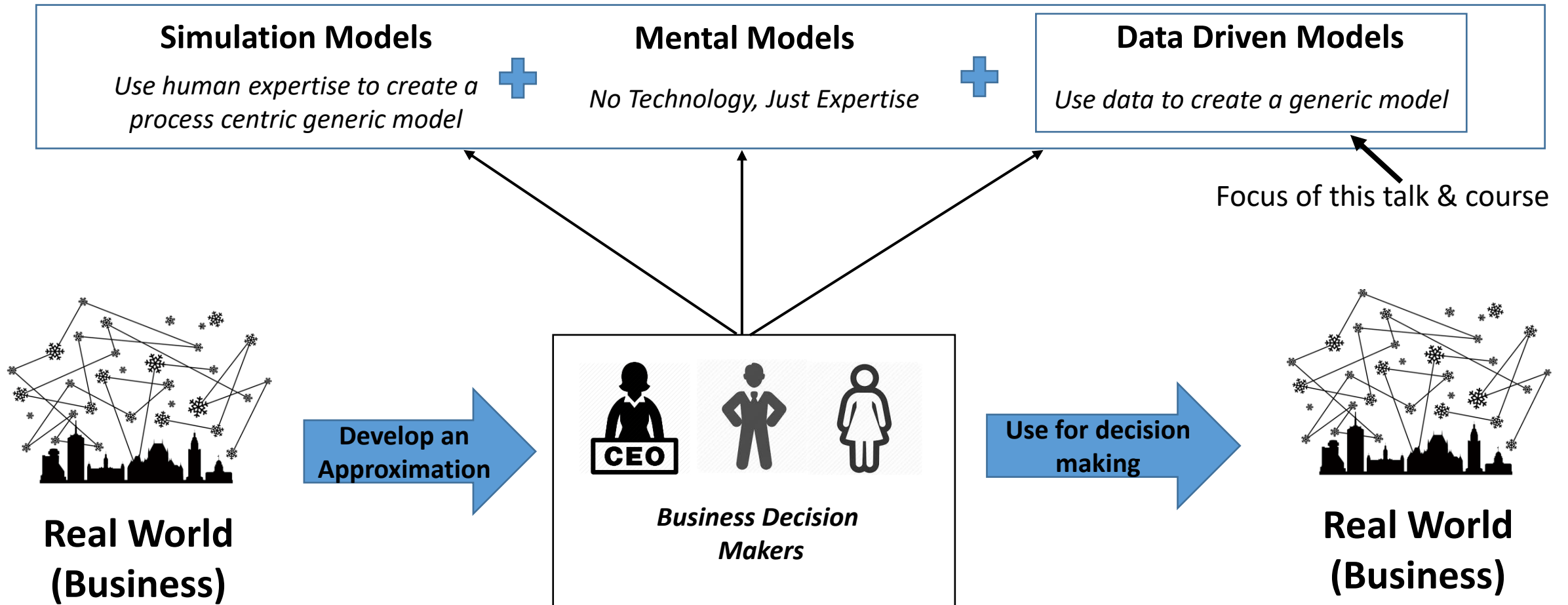


Goal & Purpose of Analytics – (with AI as Prediction Machines)

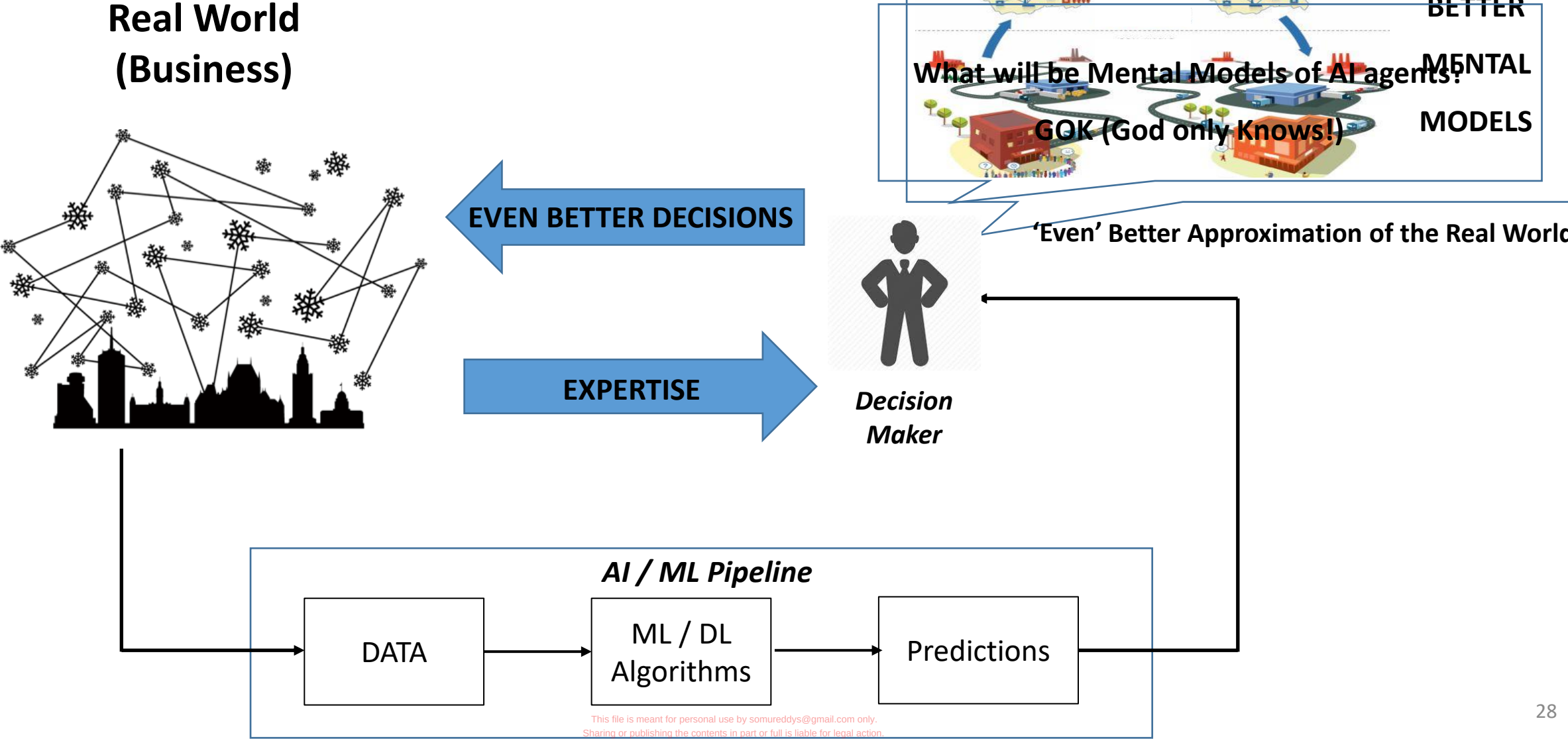


'Models of the Real World' is at the centre of business decisions

Models of the Real World



Artificial Intelligence (End to End / Autonomous)



Q6: Can you bring these concepts to life with some examples from your personal experience?



Example 1: Automobiles – Utilizing Sensor data to predict defects



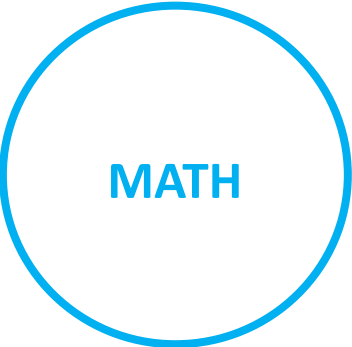
BUSINESS

For this automobile company, the warranty costs were rising year on year at an alarming rate



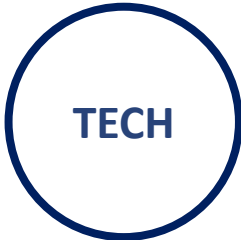
DATA

Semi-structured data from sensors from cars across 100+ countries
(Ex: Pedal position, Oil temperature, Engine temperature...60+ parameters)



MATH

Clustering done on data to identify driving styles which is then correlated with warranty claims to predict defect probability



TECH

- Spark on the Cloud platform called Databricks for Machine Learning
- User Interface using React for self-service

Example 1: Utilizing sensor data to predict defects

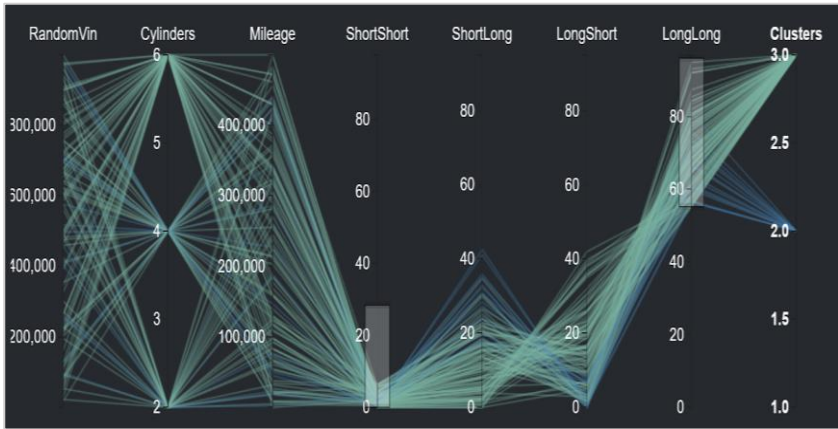
AI as Prediction Machine

Business Problem

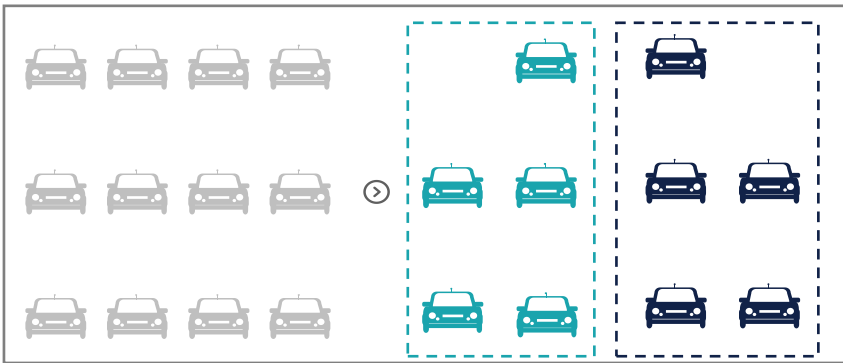
For this automobile company, the warranty costs were rising year on year at an alarming rate



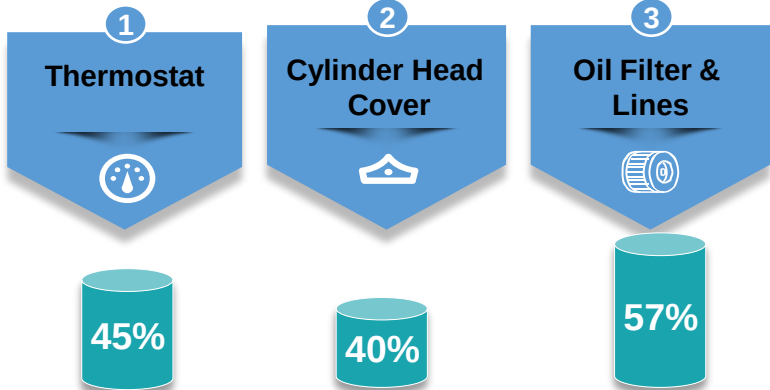
Sensor data from cars in 100+ different countries



Visualization to understand the data

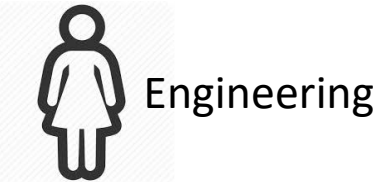


Clustering to identify driving styles



Defects were correlated with driving styles

Decisions

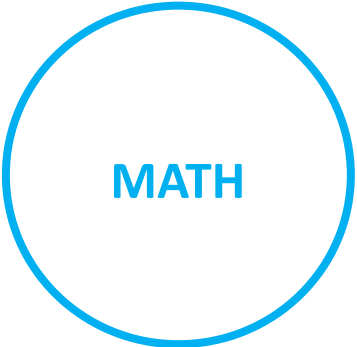


Example 2: Leveraging external data to drive innovation



BUSINESS

Consumer durables company wants to obtain product feedback as soon as its products are released in the market and not wait for 6-8 months which was the current state scenario?



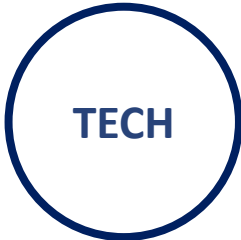
MATH

Natural Language Processing
Techniques to detect spam, emotion, entities, sentiments, contextual meaning etc.



DATA

Unstructured data from reviews in marketplaces, brand websites, Industry forums, blogs



TECH

Self-service visualization built using Tableau that provides the summary view and different levels of drill-down into specific consumer characteristics

Example 2: Using External Data to Drive Innovation

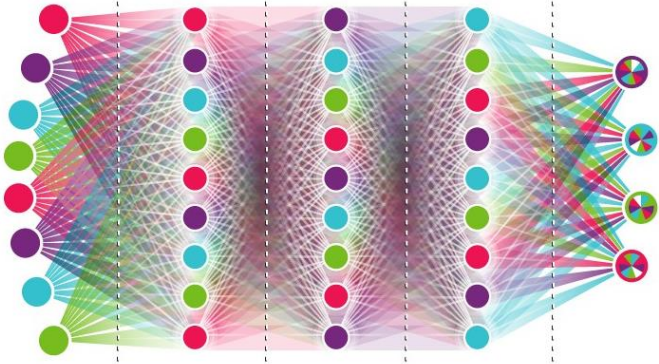
AI as Prediction Machine

Business Problem

Consumer durables company gets product feedback 6-8 months after the product is launched which is not useful



External data from ecommerce sites, brand websites etc.



Deep Learning for Text Understanding

Decisions



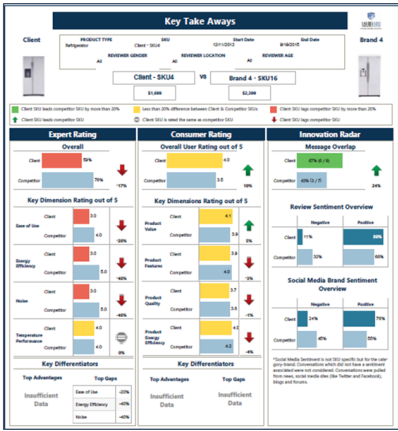
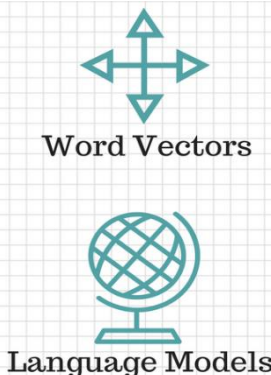
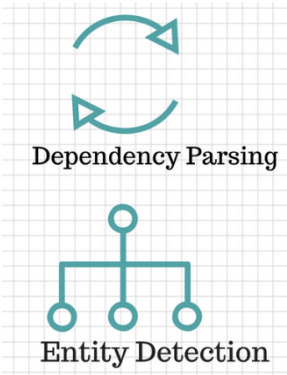
Innovation



Marketing



Distribution



Understand key factors that influence customer perception & behavior

Q7. What competencies are required to create data driven models?



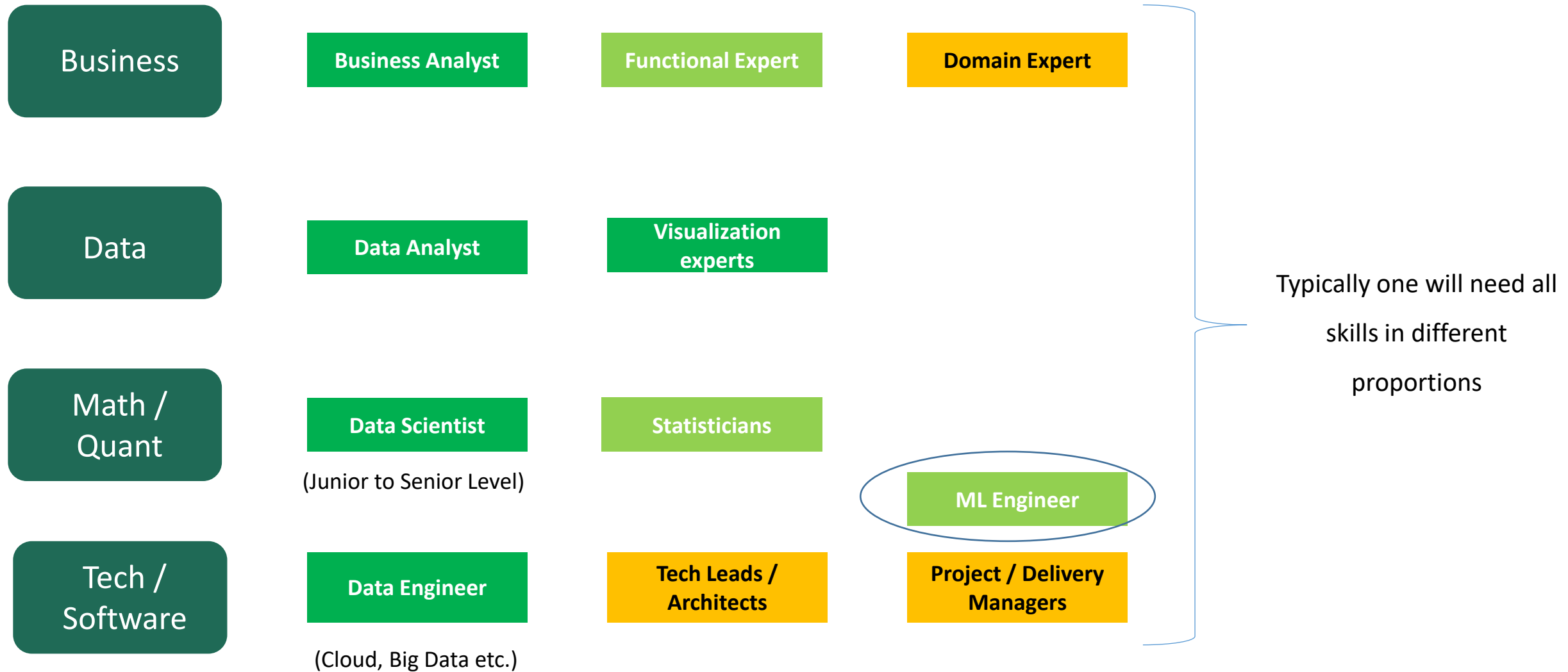
Competencies to create Data Driven Models

Business	Use Case Formulation	Interpret Analytics Output	Domain Expertise
Data	Data Types – (Un / Semi) Structured	Data Visualization & Story Telling	Signals from data (subtract noise)
Math / Quant	Select the right techniques	Appropriate coding language / platform	Evaluating the output of algos
Tech / Software	Data Engineering / Model Deployment	Front-end Applications	Software Engineering / SDLC

Q8: What are the typical roles in the analytics space and what specific skills are required to break into it?



Typical Roles in Analytics / Data Science



Entry Points for different experience levels

Fresher / Junior Developer	Lead / Architect	Program Manager / Business Head
Business Analyst	Functional / Domain Lead	Functional / Domain Expert
Data Analyst	Senior Data Analyst	Mid-Level Data Scientist
Big Data / Cloud programmer	Mid-Level Data Scientist	Analytics Delivery Manager
	Data Engineer / Big Data / Cloud Specialist	Consultant

Skills required for an Analytics Professional

- 1. Business Orientation** – Basic understanding of business concepts
- 2. Structured Thinking** – Ability to break down a problem into parts
- 3. Analytical Techniques** – Articulate as to how data & analytics can help

1. Business Orientation - Basic understanding of business concepts

Value Creation – Reason for existence of a business

Business Model – Core actors & activities that generate value

Raw material /
Ideas



Making a Product /
Offering a Service

Marketing & Selling
a Product / Service



Customers

Finance – What numbers matter & why?

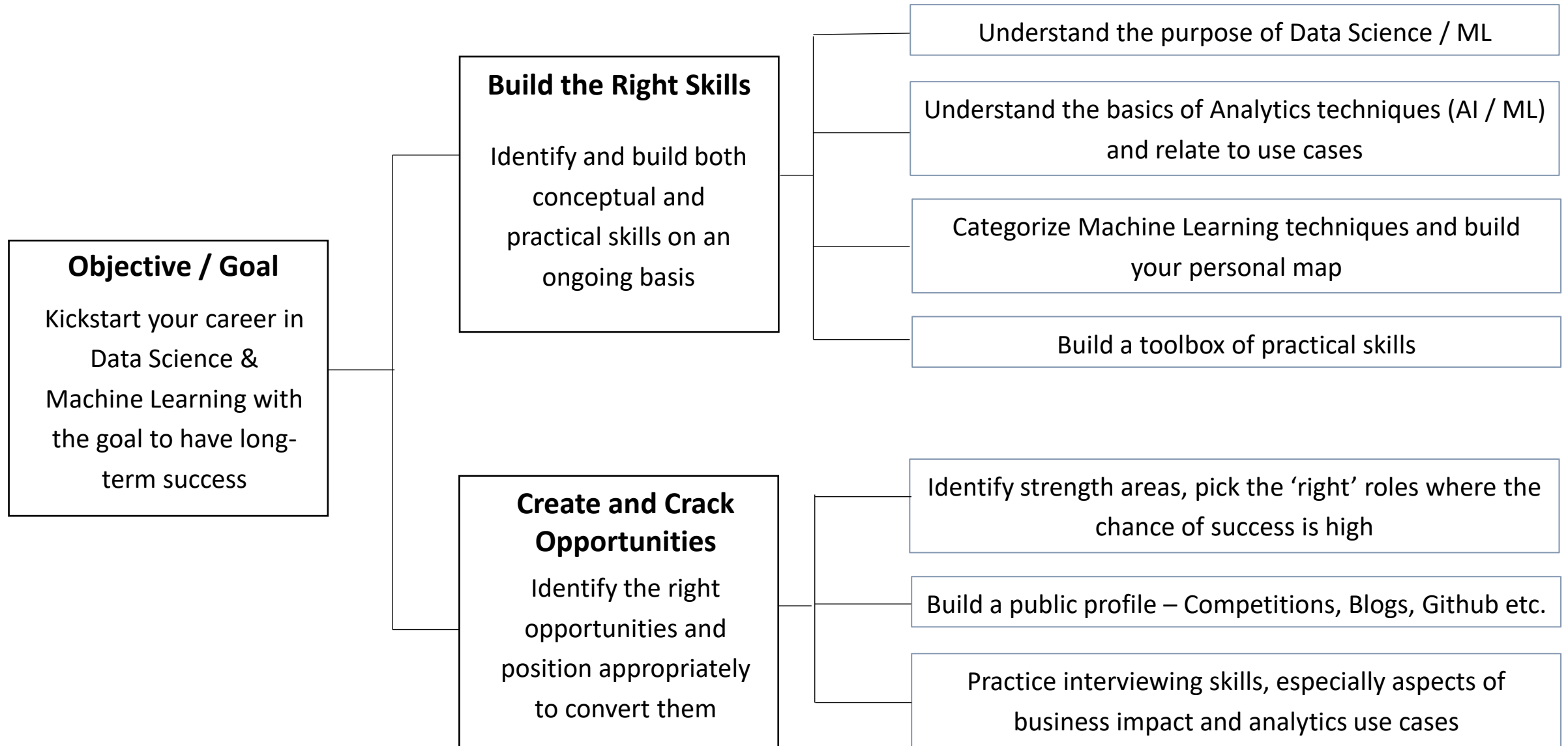
Strategy – To deal with competition

R & D – Innovation

HR – Managing People

Supporting
Functions

2. Structured Thinking – Ability to break down a problem



Structured Thinking - Many Resources are available...

Frameworks

- Cynefin (by Dave Snowden)
- Issue Tree & Hypothesis Formulation
- MECE Principle
- Porter's Five Forces
- 3C Framework (Company, Customer & Competitor)
- 4P Framework (Product, Price, Place, Promotion)
- SWOT
- 2x2 Matrices
- BCG Matrix
- McKinsey 7S Framework

Where Available

- Case Interview preparation websites like:
 - www.craftingcases.com
 - www.preplounge.com
 - Victor Cheng's CaseInterview prep
- Consulting company websites
(Ex: <http://caseinterviewprep.deloitte.com/>)
- YouTube (Case Interview Preparation videos)
- Forums like Quora

3. Analytics Techniques – Components of the toolbox

- Basic knowledge of Math concepts – Distributions, Probability, Matrices
 - Good knowledge of Statistical Techniques & Machine Learning algorithms
 - Atleast 1 programming language for ML – Python (Jupyter notebooks), R
 - Data Visualization skills – Python or Tools like Tableau, Qlikview
 - Atleast 1 GUI based ML platform – H2o, Azure ML, BigML
-
- 1 Cloud based platform (Nice to have) – AWS, Databricks, Paperspace
 - Github
 - Kaggle (Competition & Kernels), AnalyticsVidhya
 - Database / SQL knowledge (preferable)

→ *Structured course will help you get this knowledge faster*

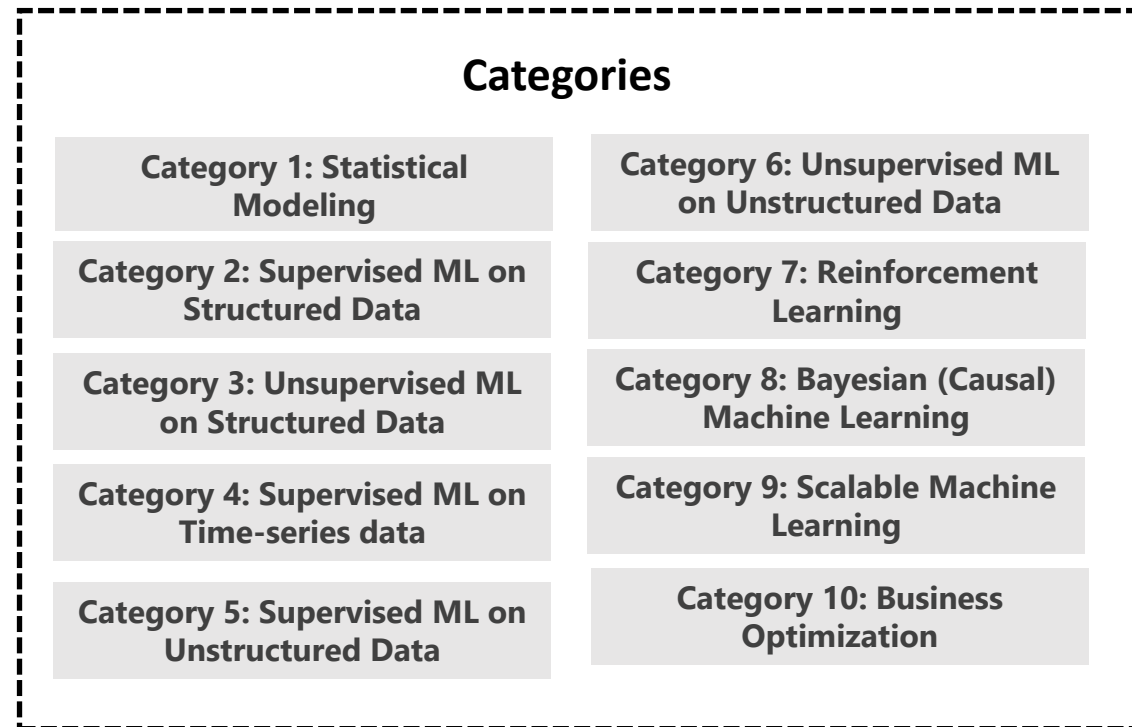
Q9: With so many algorithms & techniques available, how does one keep track and apply it?



AI / ML Techniques – There are a lot of them!



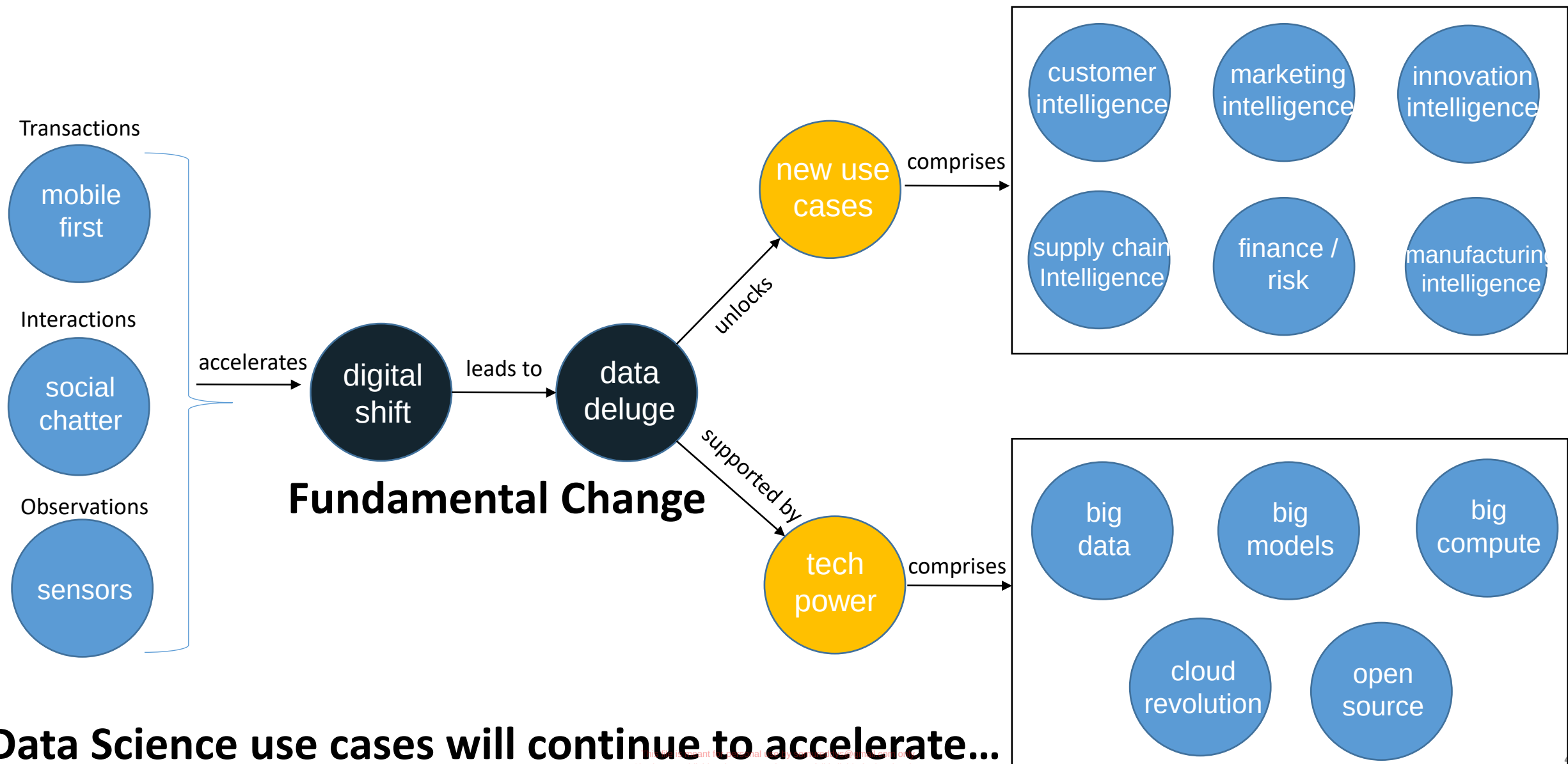
Categorization of Machine Learning Topics – My personal map



Q10: How are you sure that this field (Analytics, AI & ML) has long-term growth prospects and is not just a short-term fad?



Digital Shift is a Fundamental, Irreversible Change



Data Science use cases will continue to accelerate...

Thank You!



- Karthikeyan Sankaran, Director, LatentView Analytics
- LinkedIn – <http://in.linkedin.com/in/karthikeyansankaran>
- Mindmap – bit.ly/31KArT8